

Yuting Zhou

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EDUCATION

Rutgers University, New Jersey, United States

Sep 2024 - Expected Jun 2026

- **Major:** M.S. in Computer Science, focused on NLP, hierarchical text classification, and LLM applications.

Shanghai Institute of Technology, Shanghai, China

Sep 2020 - Jun 2024

- **Major:** B.E. in Software Engineering (GPA: 3.7/4.0, Top 10%);
- Core training in software engineering, algorithms, machine learning, and full-stack web development.

RESEARCH EXPERIENCE

Research Assistant — Hierarchical Classification, Data Engineering, LLM Applications

Jun 2025 - Present

Rutgers Center for Advanced Infrastructure and Transportation (CAIT)

New Jersey, United States

- Architected scalable data engineering pipelines to merge and reconcile 100k+ multi-source railway incident and delay records, implementing automated deduplication, hierarchical label dictionary construction, and cross-dataset consistency validation.
- Developed analytical and visualization tools to convert raw operational logs into structured daily, route-level, and location-based performance metrics, supporting delay pattern analysis and anomaly detection.
- Designed and fine-tuned hierarchical multi-label classifiers based on pretrained transformer models (e.g., DeBERTa), modeling structured dependencies with supervised fine-tuning, encoder freezing strategies, and dependency-aware evaluation.

INTERNSHIP EXPERIENCE

Front-end Engineer — Frontend Performance, Real-time Systems, Vue/React

Jun 2023 - Aug 2023

Shanghai Tenchii Digital Tech Co., Ltd. (Tencent and Shanghai Metro Joint Venture)

Shanghai, China

- Refactored rendering and request pipelines in the Shanghai Transit App, reducing page latency by 30% and improving key retention.
- Built a high-concurrency real-time lottery module with WebSocket and MongoDB, ensuring low-latency updates and stable peak-hour performance.
- Delivered 3 cross-team prototypes from Axure to Vue components with product, design, and backend teams, shortening validation cycles.

Software Test Engineer — Test Automation, CI Integration, Quality Assurance

Jun 2022 - Sep 2022

Shanghai Spacecraft Electromechanical Equipment Co., Ltd.

Shanghai, China

- Built a cross-hardware automation framework and integrated Pytest/JUnit into CI, improving regression coverage and run stability.
- Developed Selenium UI suites and regression workflows to surface high-priority defects earlier, reducing release risk and manual testing cost.
- Standardized defect severity and test reporting, helping engineers triage faster and improving release review efficiency.

PROJECT EXPERIENCE

Intelligent Rail Incident Classifier — Hierarchical Classification, DeBERTa, LLM Augmentation, Log Mining

Aug 2025 - Present

- Built a hierarchical multi-label classifier with DeBERTa + LLM augmentation to model root-cause, type, and impact dependencies in rail incident logs.
- Engineered a scalable data pipeline for multi-source logs: normalization, deduplication, hierarchy dictionary construction, and label alignment.
- Implemented an end-to-end experiment pipeline (train/search/evaluate) with reproducible configs and error-analysis reports.
- Improved structured prediction with Hierarchical Exact Match and Macro/Micro F1, reducing cross-level label conflicts.

StreamX: Movies Recommender — Hybrid Recommender, MovieLens Latest, Next.js, Django, SQLite

Feb 2026 - Present

Project Link: github.com/Magicherry/Movies-Recommendation

- Designed a hybrid recommender (MF, SVD, TextCNN) to combine collaborative and semantic signals for personalized ranking.
- On MovieLens Latest, achieved 8% lower RMSE and +11% Precision@10 versus the MF baseline.
- Built an automated evaluation pipeline (80/20 user split; MAE, RMSE, Precision@10, NDCG@10) for reproducible multi-model comparison.
- Improved diversity and coverage by 15% via TMDb metadata enrichment, and deployed a Next.js + Django service with sub-second latency.

King of Bots: Real-time Battle Platform — Distributed Session, Spring Boot, Vue, WebSocket, Redis

May 2022 - Nov 2022

Project Link: github.com/Magicherry/King-of-Bots

- Built a full-stack real-time battle platform supporting 50+ concurrent matches with <250ms end-to-end latency.
- Designed distributed session and matchmaking services with Redis cache and pub/sub for state consistency under concurrency.
- Optimized state synchronization and WebSocket multiplexing, increasing stress-test throughput by 35%.
- Built 10+ REST APIs and implemented a JWT + RBAC authorization model, improving API security and maintainability.

Bits of Me — React, UI/UX, Responsive Design, Personal Portfolio

Ongoing

Project Link: github.com/Magicherry/Magicherry.github.io

- Built a modular personal website (React 18, Router v6, Bootstrap 5) with maintainable, updatable content.
- Implemented light/dark themes, glassmorphism, animations, mobile nav, filtering, real-time PDF viewer, and GitHub visualization.

CERTIFICATES & SKILLS

- **Certifications:** NVIDIA Transformer-Based NLP, NVIDIA Intro to Deep Learning, Alibaba Cloud ACA (Big Data), Huawei HCIA (Intelligent Computing).
- **Repository:** github.com/Magicherry (including machine learning, real-time systems, and full-stack web applications).
- **Programming:** Python, C++, Java, Go; **Data & ML:** PyTorch, Transformers, scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn.
- **Engineering:** Mixed-precision, gradient accumulation, multi-GPU/MPS training; **Deployment:** RESTful APIs, Docker, Kubernetes, GitHub Actions.
- **Frontend:** Vue 3, React, Next.js; **Backend:** Django, FastAPI, Spring Boot, Redis, MongoDB, MySQL;
- **Languages:** English, Mandarin Chinese (Bilingual).